

SwingPortfolio

Multi-Strategy Swing Trading System for NSE F&O Stocks
Documentation - June 2026

1. Overview

SwingPortfolio is a modular, configuration-driven swing trading system for Indian NSE F&O stocks. It runs multiple independent strategies from a single codebase with shared data loading, backtesting, live trading, Google Sheets sync, and dashboard monitoring.

Active Strategies

- Donchian ADX -- 20-day breakout + ADX > 25 + DI confirmation (Enabled)
- Keltner RSI + Volume -- Keltner bands + RSI 55/45 + Volume 1.5x avg (Enabled)
- Ichimoku Cloud -- TK Cross + Kumo + ADX > 25 (Disabled -- pending maturity)

Capital & Position Sizing

Parameter	Value	Description
Total Capital	Rs. 500,000	Across all strategies
Margin/Position	~Rs. 200K + 50K buffer	Lot size x price x 15% + buffer
Max/Strategy	2	Max open positions per strategy
Max Total	4	Max across both strategies (2x2)
Position Sizing	Lot-based	Uses actual NSE F&O lot sizes
Universe	Margin < 200K	Dynamically filtered from yfinance

Risk Parameters

Parameter	Value	Description
SL_ATR_MULTIPLIER	3.0	Stop = ATR x 3 (was 2.0)
TRAIL_PCT	3%	Trailing from best price (was 0.75%)
HARD_SL_AMOUNT	Rs. 75,000	Max loss per position (was 25K)
RR_RATIO	1.5	Risk-reward target
GAP_FILTER	Enabled	Skips entry if gap > 3%

Run Schedule

Time	Task	Description
08:30	Stock Selection	Scan yesterday's close for signals
09:15	Market opens	
09:42	Entry Phase	Read selections, zscore entry on 1-min data
09:42-15:15	Monitor Phase	Every 60s: check SL/trail, retry unfilled
15:15	End of Session	OPEN positions carry forward
15:30	Market closes	

Entry Methods

Mode	Description
zscore	Enter when z-score exceeds 2s from 15-min baseline. Retries every 5 min.
walk	Enter when price breaks outside first 15 candles' high/low range
15m_945	Fixed entry at 09:45 using 15-min candle open

2. Architecture

```
SwingPortfolio/
|
+-- cfg/                                # Configuration
|   +-- settings.py                    #   Global params (capital, margin, risk, SL)
|   +-- universes.py                  #   Symbol lists, lot sizes, margin filter
|   +-- strategies.json                #   Strategy registry
|
+-- core/                               # Shared engine
|   +-- strategy_base.py              #   Abstract base class
|   +-- registry.py                   #   Auto-discovers strategies
|   +-- data_loader.py                #   yfinance daily downloader
|   +-- backtest_engine.py            #   Backtest + simulate_today()
|   +-- google_sheets.py              #   Push/pull sync (5 tabs)
|   +-- atomic.py                     #   Atomic file I/O
|   +-- notifications.py              #   Telegram alerts
|
+-- strategies/                         # Strategy logic
|   +-- donchian_adx.py                #   Donchian 20d + ADX>25
|   +-- keltner_rsi.py                 #   Keltner + RSI + Volume
|   +-- ichimoku.py                   #   Ichimoku (disabled)
|
+-- agents/                             # Live agents (Scheduled Tasks)
|   +-- stock_selection.py             #   08:30 scan
|   +-- live_trader.py                 #   09:42 entry + 60s monitor
|   +-- sheet_watcher.py              #   Google Sheets sync
|
+-- reports/                            # Streamlit dashboard
|   +-- dashboard.py                  #   Grouped positions, color-coded P&L
|   +-- backtest.py                   #   Backtest + Simulate Today
|   +-- stock_selection.py             #   Scan results
|   +-- admin.py                      #   Settings, Telegram, Sheet
|   +-- scheduler.py                  #   Task Scheduler
|   +-- about.py                      #   This documentation
|
+-- data/                               # Runtime files
|   +-- selections.json                #   Pre-market scan
|   +-- live_positions.json            #   Open/closed positions
|   +-- positions_history.json         #   All-time closed
|   +-- signals.csv                    #   Signal log
|   +-- margin_universe.json           #   Filtered universe cache
|   +-- sheet_config.json              #   Google Sheet config
|   +-- google_service_key.json        #   Service Account key
|   +-- telegram_config.json           #   Bot token + chat ID
|
+-- cache/                              # Pickled daily OHLCV
|   +-- daily_YYYYMM.pkl              #   One file per month
|
+-- dashboard.py                       # Streamlit entry point
```

3. Configuration

cfg/settings.py - Global Parameters

Parameter	Current	Description
CAPITAL_TOTAL	500,000	Total capital across all strategies
MAX_POSITIONS_PER_STRATEGY	2	Max concurrent positions per strategy
MAX_POSITIONS_TOTAL	4	Max total positions across both
POSITION_SIZING	lot	Uses NSE F&O lot sizes
MAX_MARGIN_PER_STOCK	200,000	Max futures margin per position
MARGIN_RATE	0.15	Margin as % of contract value
TRAIL_PCT	0.03 (3%)	Trailing SL from best price
SL_ATR_MULTIPLIER	3.0	Stop = ATR x 3
RR_RATIO	1.5	Risk-reward target
HARD_SL_AMOUNT	75,000	Max loss per position
HARD_SL_ENABLED	True	Hard SL active
GAP_FILTER_ENABLED	True	Skip entry if gap > 3%
ENTRY_TIME	zscore	Entry method

Why These Values?

- SL_ATR_MULTIPLIER = 3: 1.5x the daily range as a stop. Wide enough to avoid intraday noise.
- TRAIL_PCT = 3%: Swing trades need room. Old 0.75% caused premature exits.
- HARD_SL = 75K: Increased from 25K so ATR-based SL is the primary exit mechanism.

Strategy Registry (cfg/strategies.json)

Each strategy is a JSON entry with fields: id, name, module, class, description, enabled, universe, entry_time, and params. Setting enabled = false skips the strategy in all agents. The universe field selects which stock pool to scan (Nifty 50, Nifty 100, Full F&O, Margin < 200K).

Universe Filtering (cfg/universes.py)

SYMBOLS_DATA contains 215+ NSE F&O symbols with lot sizes. The Margin < 200K preset dynamically filters out any stock where $\text{lot_size} \times \text{price} \times 0.15 > 200,000$. Results are cached in data/margin_universe.json and refreshed if older than 1 hour.

4. Strategies

4.1 Donchian + ADX (Enabled)

Concept: Breakout trading. Enters when price breaks out of a 20-day consolidation range, confirmed by trend strength ($ADX > 25$) and direction (DI crossover).

Entry LONG: $Close > 20\text{-day high AND } ADX > 25 \text{ AND } +DI > -DI$

Entry SHORT: $Close < 20\text{-day low AND } ADX > 25 \text{ AND } -DI > +DI$

Exit: $SL = \text{reference} - ATR \times 3$ (LONG) or $+ ATR \times 3$ (SHORT). Trail = 3% from best price. Target = 1:1.5 RR. Hard SL = Rs. 75K max loss.

4.2 Keltner + RSI + Volume (Enabled)

Concept: Momentum entry. Enters when price pushes beyond Keltner bands with strong RSI momentum and above-average volume confirming the move.

Entry LONG: $Close > \text{upper band AND } RSI > 55 \text{ AND } Vol > 1.5 \times 20\text{-day avg}$

Entry SHORT: $Close < \text{lower band AND } RSI < 45 \text{ AND } Vol > 1.5 \times 20\text{-day avg}$

Exit: Same as Donchian ($ATR \times 3$ SL, 3% trail, 1:1.5 RR, Rs. 75K hard SL).

4.3 How They Complement Each Other

Donchian catches breakouts (trending moves after consolidation), while Keltner catches momentum with volume (strong directional pushes). Their losing patterns rarely overlap because they respond to different market micro-structures.

- 73% of weeks both strategies profit simultaneously
- 23% of weeks one wins while the other loses (net positive)
- Only 3% of weeks (1 of 30) both lose simultaneously

4.4 Adding a New Strategy

1. Create a class in strategies/ extending BaseStrategy with prepare_data() and check_signal()
2. Add a JSON entry in cfg/strategies.json with id, module, class, params
3. Restart the dashboard -- it auto-discovers the new strategy

5. Backtest Results

5.1 Donchian ADX -- Month-wise (Nov 2025 - May 2026)

Month	Trades	Wins	Loss	WR%	P&L
Nov	16	8	8	50%	+84,009
Dec	19	13	6	68%	+331,828
Jan	22	15	7	68%	+109,588
Feb	11	8	3	73%	+82,169
Mar	21	17	4	81%	+221,290
Apr	28	25	3	89%	+538,388
May	25	18	7	72%	+250,081
Total	142	104	38	73%	+1,617,353

5.2 Keltner RSI -- Month-wise (Nov 2025 - May 2026)

Month	Trades	Wins	Loss	WR%	P&L
Nov	16	12	4	75%	+248,464
Dec	17	15	2	88%	+133,163
Jan	21	16	5	76%	+170,094
Feb	18	13	5	72%	+186,609
Mar	30	22	8	73%	+361,210
Apr	10	8	2	80%	+63,307
May	14	7	7	50%	+54,950
Total	126	93	33	74%	+1,217,797

Note: With max 2 position constraint (Rs. 500K capital), the backtest scales to 70 trades / +721,364 P&L for Donchian and 62 trades / +694,521 P&L for Keltner RSI.

5.3 Combined Portfolio Summary

Metric	Donchian	Keltner	Combined
Trades	70	62	132
Win Rate	69%	76%	72%
P&L	+721,364	+694,521	+1,415,885
Max DD	Rs. 66K (13%)	Rs. 104K (21%)	~Rs. 85K (17%)
Avg Hold	4 days	4 days	4 days

5.4 June 2026 Trade Log -- Donchian ADX

Date	Sym	Dir	Entry	Exit	P&L	Hold
01-Jun	BRITANNIA	SHORT	5174	5212	-4,750	8d
01-Jun	PNBHOUSING	SHORT	1031	990	+26,889	2d
01-Jun	RVNL	SHORT	246	240	+12,143	1d
01-Jun	NESTLEIND	SHORT	1407	1409	-1,321	2d
03-Jun	IDEA	LONG	14	15	+40,919	5d

04-Jun	LAURUSLABS	LONG	1420	1413	-5,704	4d
04-Jun	SAMMAANCAP	LONG	179	187	+31,131	1d
08-Jun	RVNL	SHORT	231	231	+347	1d
09-Jun	POLICYBZR	SHORT	1518	1529	-3,780	0d
09-Jun	ADANIGREEN	LONG	1515	1506	-5,340	0d

5.5 June 2026 Trade Log -- Keltner RSI

Date	Sym	Dir	Entry	Exit	P&L	Hold
01-Jun	BDL	SHORT	1198	1217	-7,964	2d
01-Jun	COLPAL	SHORT	2008	2085	-20,955	8d
02-Jun	FORCEMOT	SHORT	17888	17911	-567	1d
02-Jun	APLAPOLLO	SHORT	1788	1803	-5,147	1d
05-Jun	POLICYBZR	SHORT	1550	1523	+9,468	4d
05-Jun	HINDZINC	SHORT	595	571	+30,111	4d
05-Jun	PATANJALI	SHORT	426	427	-1,021	4d

6. Deployment

6.1 Scheduled Tasks (Windows Task Scheduler)

Task	Trigger	Duration	Description
Stock Selection	08:30	Instant	Scan yesterday's close, save selections
Live Trader	09:35	to 15:15	Zscore entry, 60s monitor loop
Sheet Watcher	09:15	to 15:30	Google Sheets sync every 60s
Streamlit App	09:00	Continuous	Dashboard on localhost:8501

All tasks run Monday-Friday (NSE trading days) with interactive logon. Auto-restart on failure (up to 3 attempts). Managed from the Scheduler page.

6.2 Google Sheets Integration

The system syncs via a service account with 5 auto-created tabs:

Tab	Direction	Content
Control	Sheet -> Python	Enable/disable strategies, close positions
Positions	Python -> Sheet	Open + closed positions (grouped by strategy)
History	Python -> Sheet	All-time closed positions
Status	Python -> Sheet	Sync timestamp, open count, health
Selections	Python -> Sheet	Latest pre-market scan results

Control commands: Set `donchian_adx` / `keltner_rsi` = ENABLED or DISABLED to toggle strategies remotely. Set `close_position` = SYMBOL_STRATEGY (e.g., `CONCOR_donchian_adx`) to force close.

6.3 First-Time Setup

```
cd C:\claudecode\SwingPortfolio
pip install streamlit pandas yfinance plotly requests numpy gspread oauth2client fpdf2
streamlit run dashboard.py
```

1. Admin > Risk Settings: verify params, click Refresh from yfinance
2. Admin > Telegram: enter bot token + chat ID from @BotFather
3. Admin > Google Sheet: enter Sheet ID, enable, Test Sync